



Water Monitoring and Truck Tracking Web System

Vibe Tech Requirements Specification:

1.Nosipho Mbatha 202406965

2.Nomcebo Nkosi 202402024

3 Sisekelo Masombuka 202448576

4.Jabulile Shabalala 202324579

HEADINGS	PAGES
BUSINESS PROCESSES	3-4
NON-FUNCTIONAL REQUIREMENTS	5
FUNCTIONAL REQUIREMENTS	6-7
UML DIAGRAMS	7-16
PROJECT MANAGEMENT ARTEFACTS	17

TABLE OF CONTENT

Water Monitoring and Truck Tracking Web System

BUSINESS PROCESSES

User Registration and Login

Resident registers on the system: Resident visits the web app > clicks "Register" > enters name, surname, area/location, email and password > system validates inputs > system creates account and stores credentials in User Database > resident receives confirmation email > resident logs in with credentials > system authenticates > resident is directed to Resident Dashboard.

Admin logs in: Municipality admin navigates to admin portal > enters email and password > system verifies credentials against User Database > system grants access to Admin Dashboard.

Viewing Dam Water Levels

Resident logs into system > navigates to "Dam Levels" section > system retrieves real-time and historical dam level data from Water Data Repository > system displays current dam level percentage and historical trend graph > resident views water availability status and conservation tips.

Water Shortage Alert and Notification

System continuously monitors dam level data > dam level drops below defined threshold (e.g. below 30%) > Water Monitoring and Analytics module triggers alert > Notification Engine sends SMS via SMPP Gateway and email via SMTP to all registered residents in affected areas > residents receive early warning notification with conservation advice and expected restriction dates.

Viewing Water Restrictions Schedule

Resident logs in > navigates to "Water Restrictions" section > system retrieves scheduled restriction data from Water Data Repository > system displays restriction schedule by area and date > resident views their area's restriction times and plans accordingly.

Real-Time Water Truck Tracking

Resident logs in > navigates to "Track Trucks" section > system retrieves simulated GPS location data from GPS/Mapping Simulation API > system displays truck locations on interactive map > resident sees which truck is nearest to their area > resident views estimated delivery time and delivery schedule for their area.

Admin Managing Water Truck Deliveries

Municipality admin logs in > navigates to "Truck Management" section > admin adds or updates truck details in Logistics Database > admin assigns delivery routes and schedules to drivers > system saves updated schedule > drivers receive their assigned routes > system tracks delivery progress > admin monitors deliveries through Admin Dashboard > system generates delivery performance reports.

Resident Submitting a Complaint

Resident logs in > navigates to "Complaints" section > resident fills in complaint form (missed delivery, water quality, etc.) > system validates and stores complaint in Feedback Database > system assigns a complaint reference number > resident receives confirmation > admin views complaint on dashboard > admin updates complaint resolution status > system notifies resident when complaint is resolved.

Admin Updating Dam Data and Generating Reports

Admin logs in > navigates to "Water Monitoring" section > admin manually updates dam level data or uploads historical data file > system stores updated data in Water Data Repository > system runs risk analysis based on updated data > system generates monthly **distribution report** > **admin** views high-risk zones on analytics dashboard > admin uses report to plan truck deployments.

NON-FUNCTIONAL REQUIREMENTS

1. Performance

- Pages must load and respond in under 3 seconds.
- Live updates for water levels and truck locations must appear within 5 seconds.
- The system must stay fast even with many users online at once.

2. Security

- User passwords must be stored with encryption.
- Access must require user authentication.
- Different roles (Resident, Admin, Driver) must have role-based permissions.
- Inactive sessions must end with automatic logout.

3. Reliability & Availability

- The system must run 24/7 with little to no downtime.
- Regular backups must be used to avoid data loss.
- All complaints, reports, and water records must be stored securely.

4. Usability

- The interface must be simple and easy for all users to understand.
- The design must be responsive and work well on mobile devices.
- All messages and prompts must be clear and direct.

5. Scalability

- The system must support adding more users, trucks, and dam sites in future.
- It must manage growing amounts of data without needing a full rebuild.

6. Maintainability

- Updates and fixes must be simple to apply.
- Admins must be able to manage data without needing developer help.

7. Data Accuracy

- Stored data must always be correct and current. Any uploaded data must be checked and validated before it's saved.

8. Compatibility:

- The system must run in major browsers such as Chrome, Edge, and Firefox. It must function properly on both mobile and desktop devices.

FUNCTIONAL REQUIREMENTS

1. User Management (Residents, Admins, Drivers)

- The system shall allow users to register accounts.
- The system shall allow users to log in safely.
- The system shall support different user roles (Resident, Admin, Driver).
- The system shall allow users to manage and update their profiles.
- The system shall authenticate users before giving them access.

2. Water Monitoring System

- The system shall show real-time dam water levels.
- The system shall show past water level trends.
- The system shall allow admins to update the data of dam water.
- The system shall allow admins to upload past data.
- The system shall create water shortage risk analysis.
- The system shall create monthly water allocation reports.

3. Water Truck Tracking System

- The system shall display real-time truck locations (GPS simulation).
- The system shall allow admins to manage water trucks.
- The system shall allow admins to allocate delivery routes.
- The system shall allow admins to plan deliveries.
- The system shall allow users to view delivery schedules by area.
- The system shall allow admins to track the progress of deliveries.
- The system shall track the records of driver performance.

4. Notification and Alert System

- The system shall create early warning alerts for water shortages.
- The system shall send SMS notifications to residents.
- The system shall send email notifications to residents.
- The system shall generate alerts based on predictive data analysis.

5. Water Restrictions & Information Sharing

- The system shall show planned water restrictions.
- The system shall provide water conservation tips.
- The system shall allow residents to view important announcements.

6. Complaint Management System

- The system shall allow residents to submit complaints.
- The system shall allow residents to track their complaint status.

- The system shall allow admins to view and manage complaints.
- The system shall allow admins to update complaint resolution status.

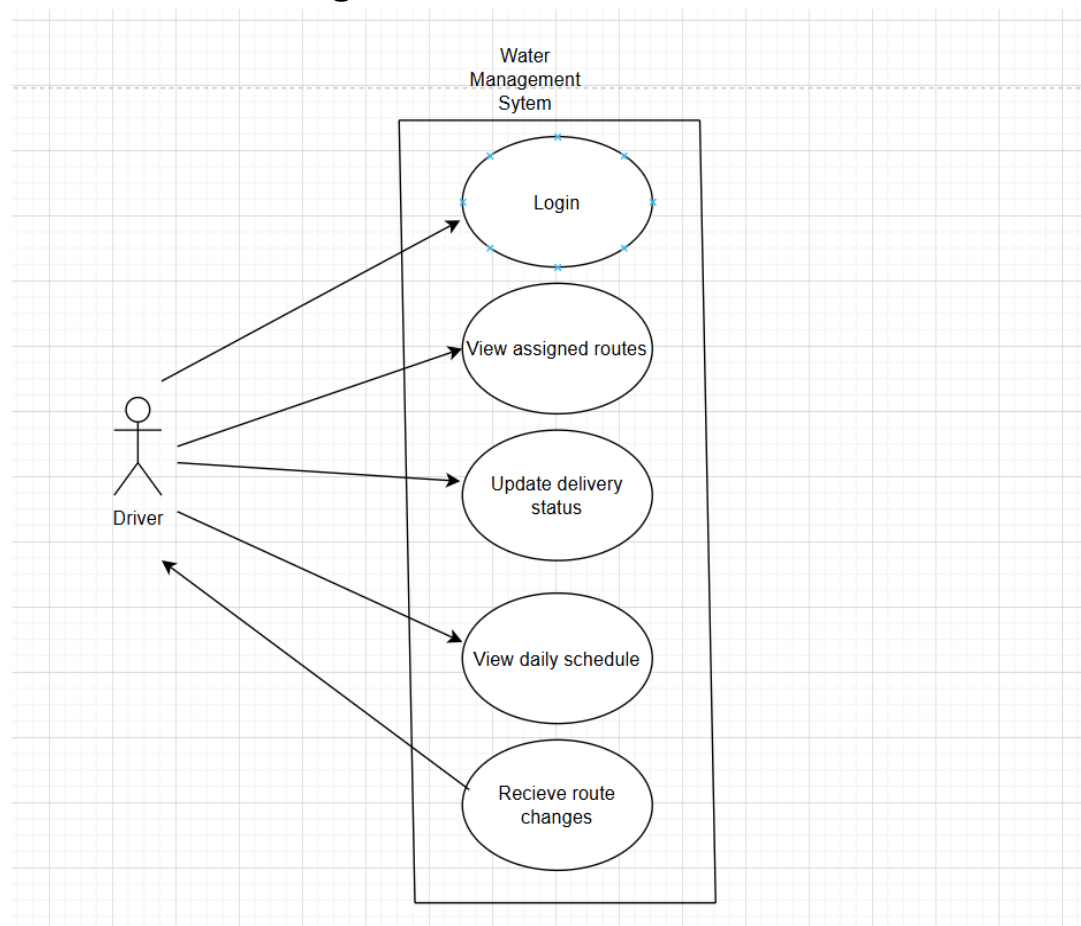
7.Admin Dashboard & Analytics

- The system shall provide an analytics dashboard.
- The system shall show high-risk zones.
- The system shall allow admins to monitor the performance of the system.
- The system shall generate reports for decision-making.

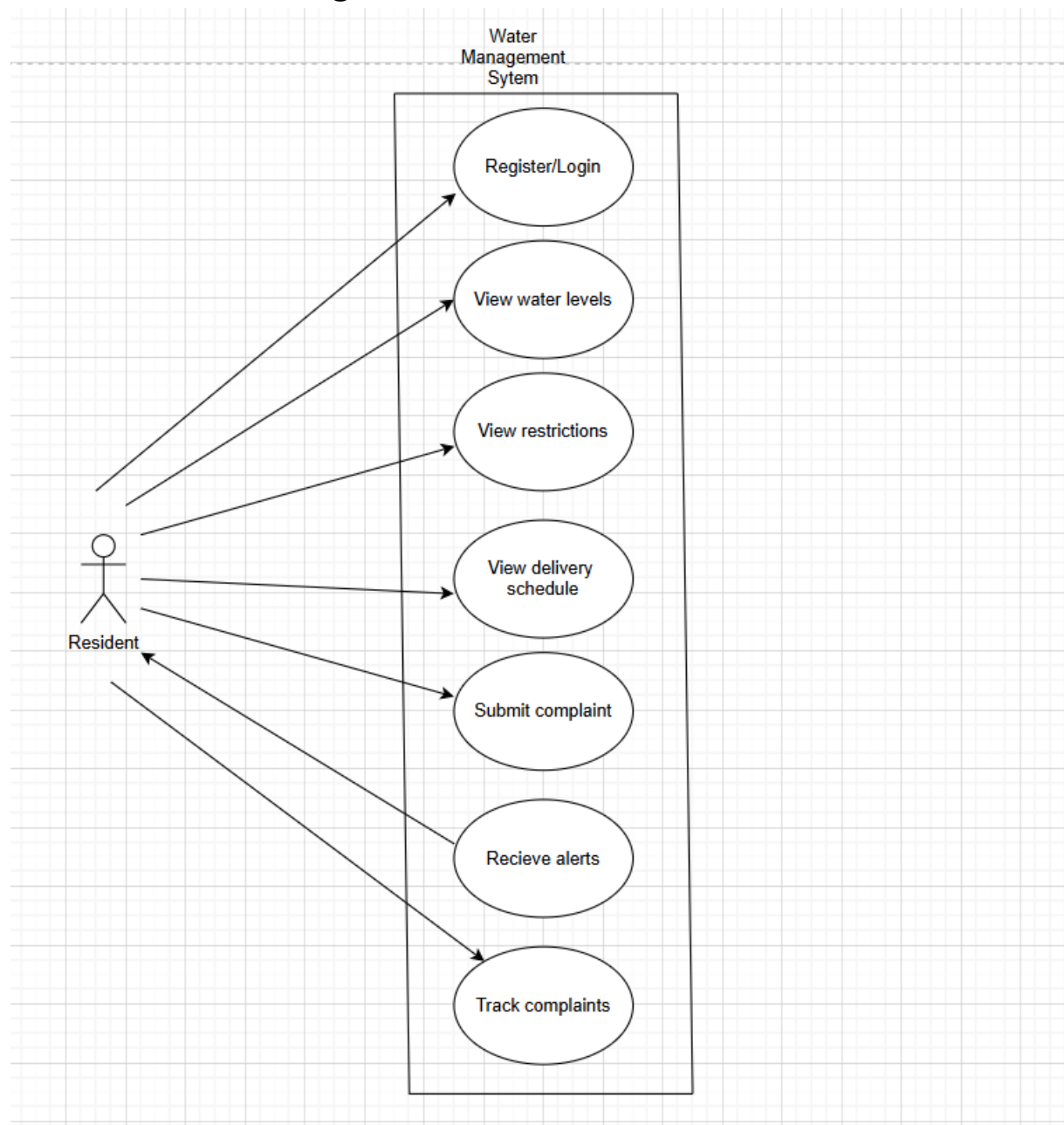
UML DIAGRAMS

Use case diagrams

Driver use case diagram



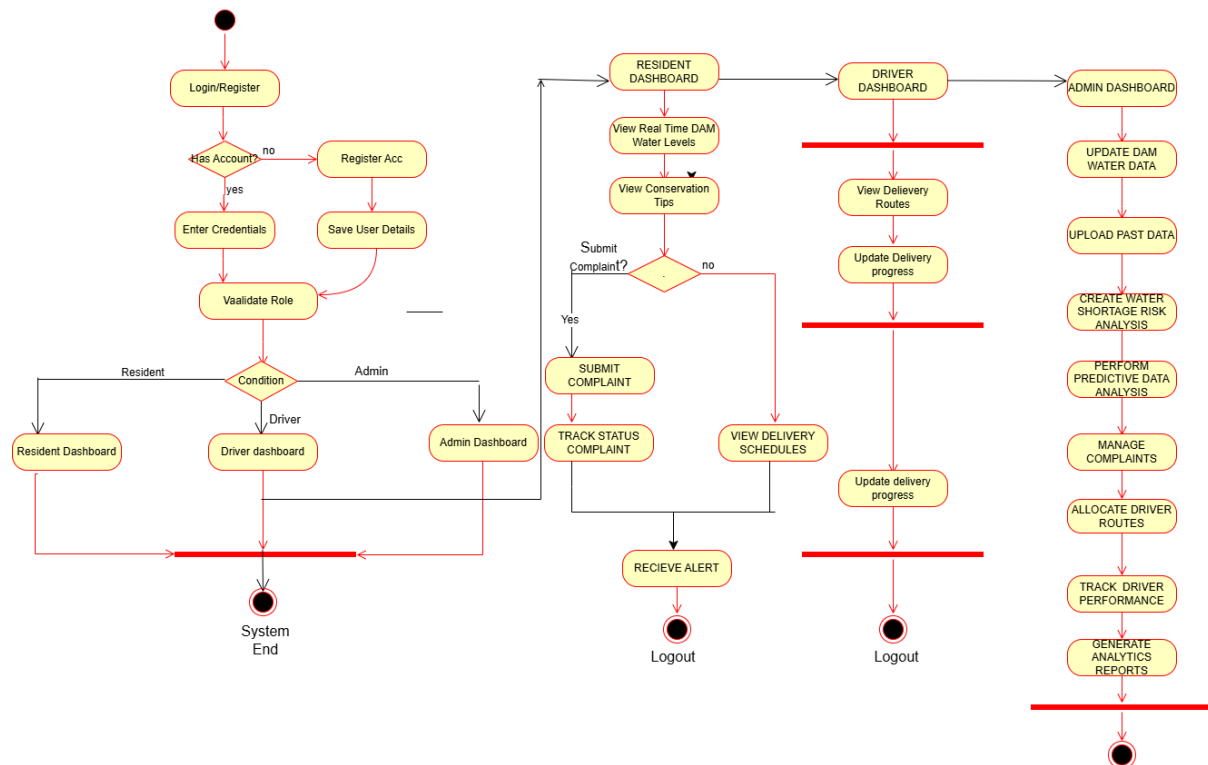
Resident use case diagram



Admin use case diagram

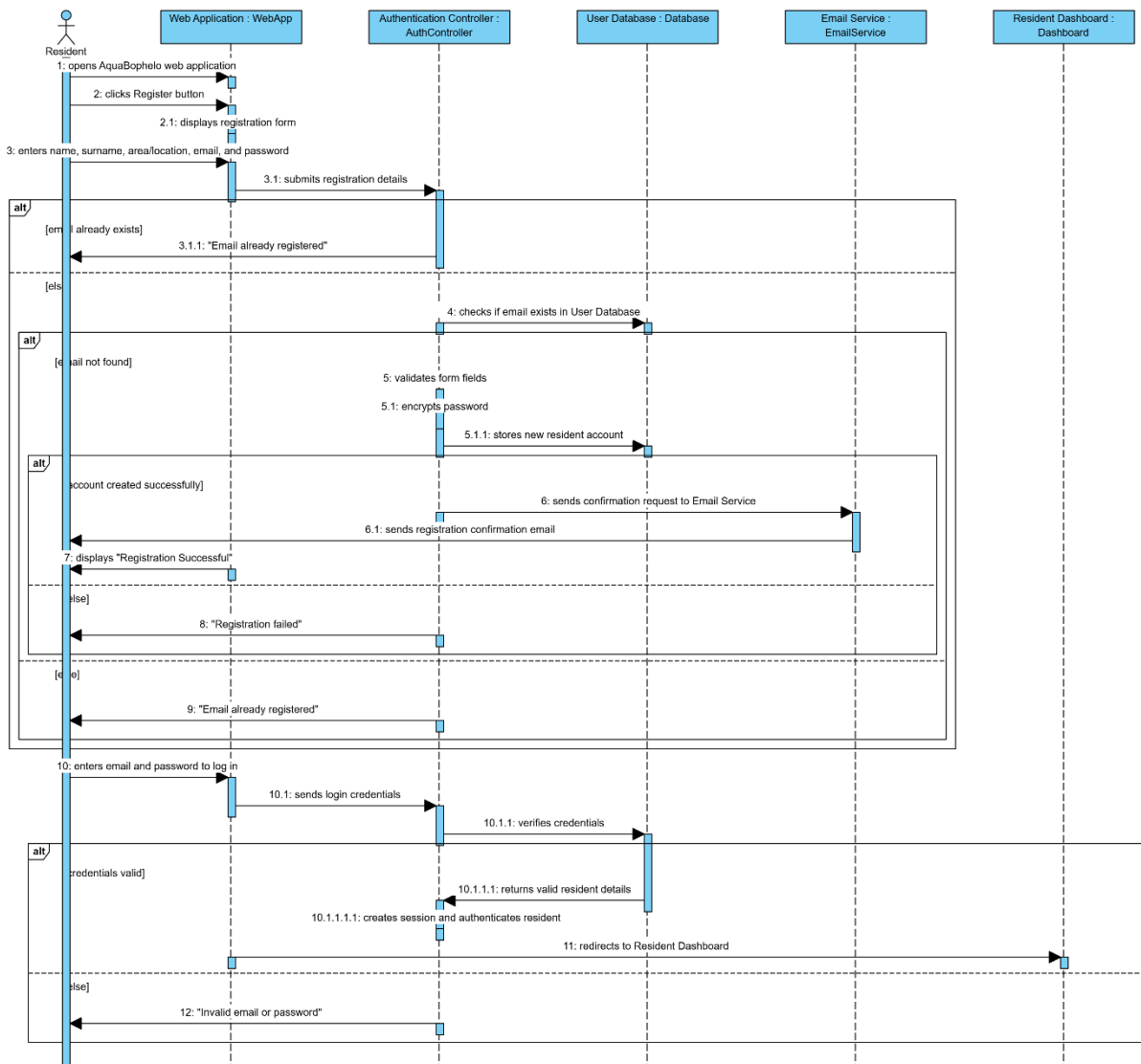


Activity diagram

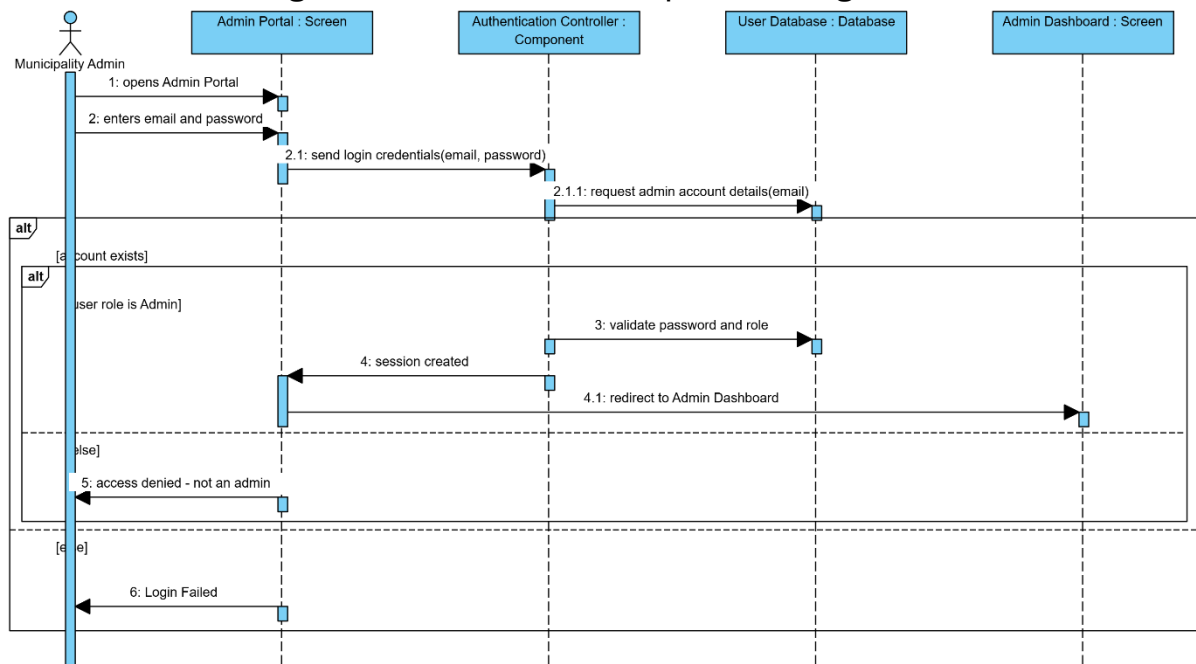


Sequence diagrams

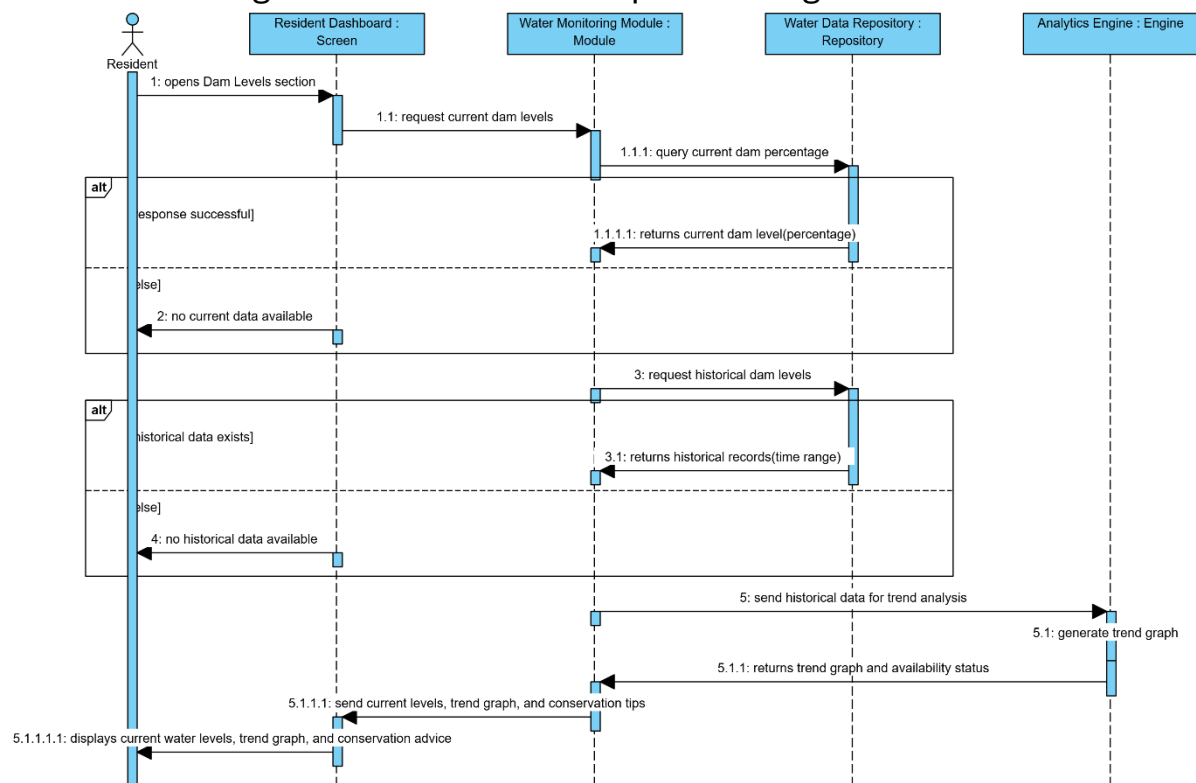
1. Resident Registration and Login Sequence Diagram:



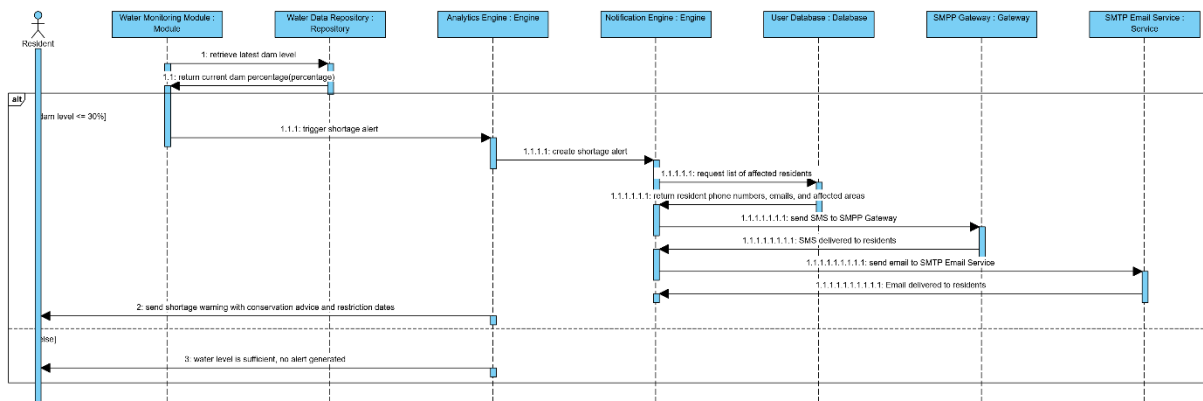
2. Resident Viewing Dam Water Levels Sequence Diagram:



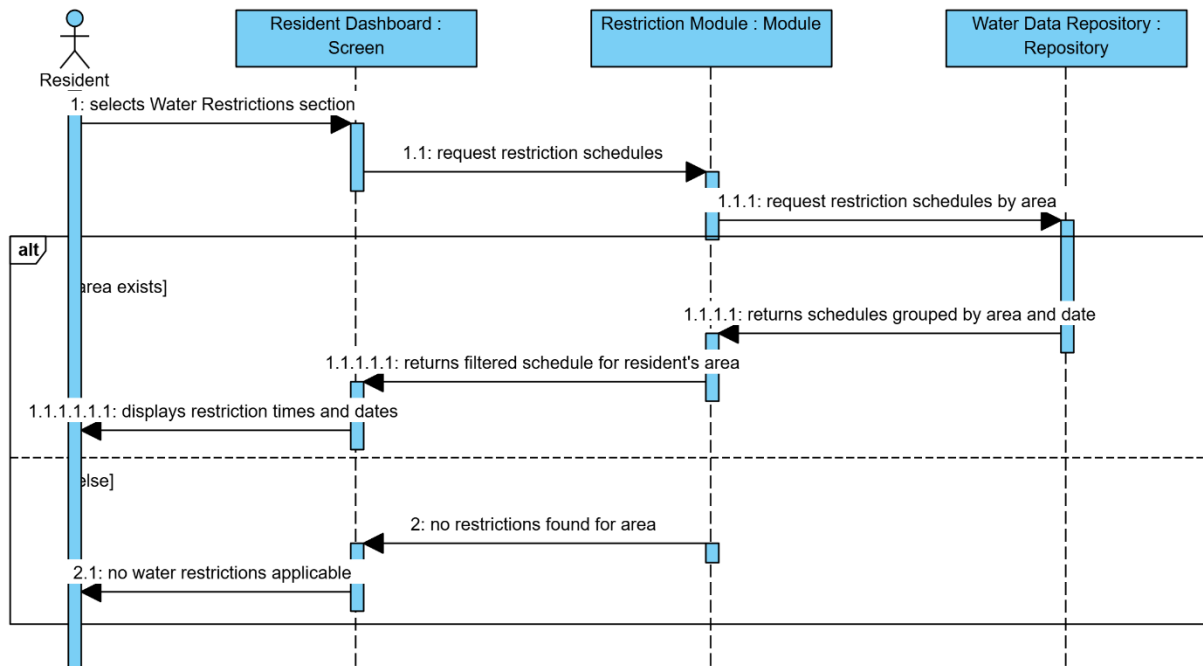
3. Water Shortage Alert Notification Sequence Diagram:



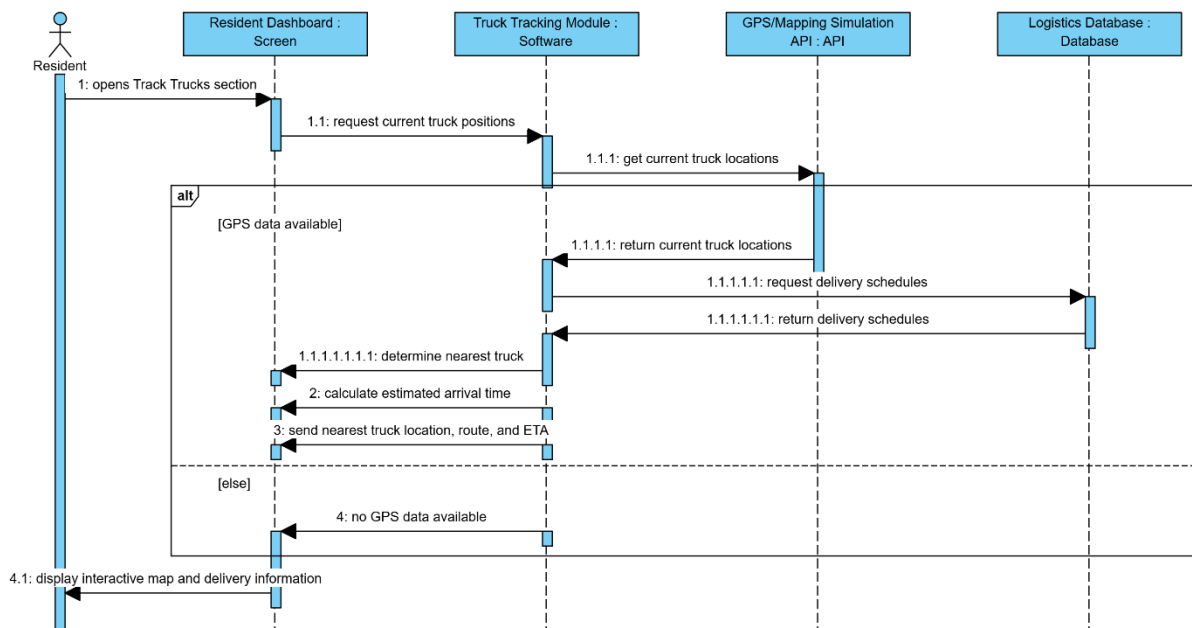
4. Resident Viewing Water Restrictions Sequence Diagram:



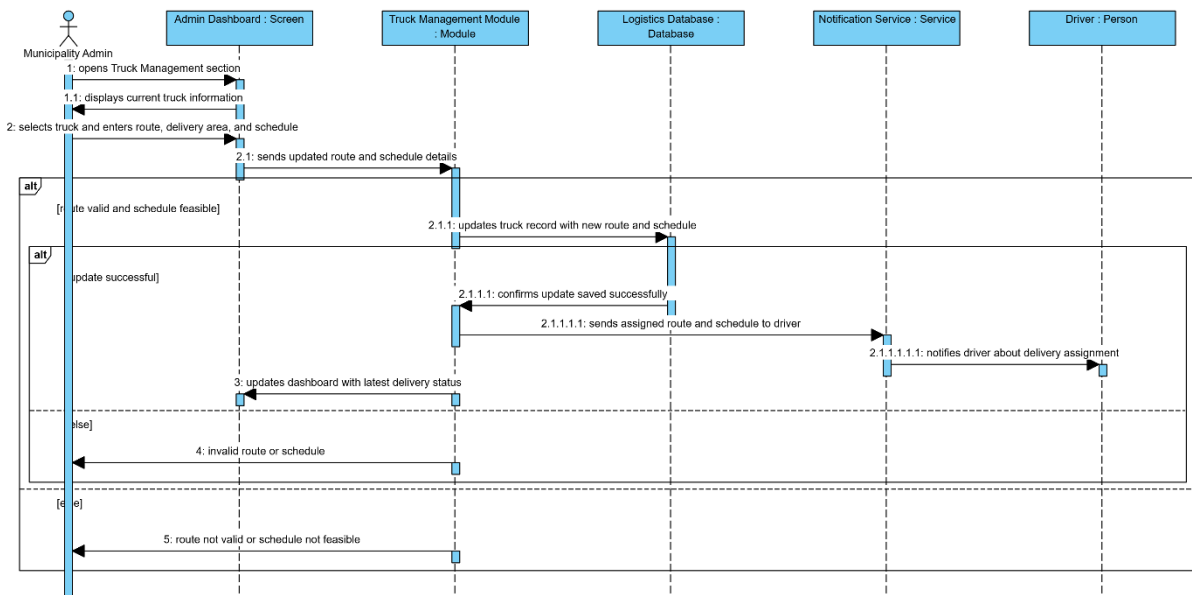
5. Resident Tracking Water Trucks Sequence Diagram:



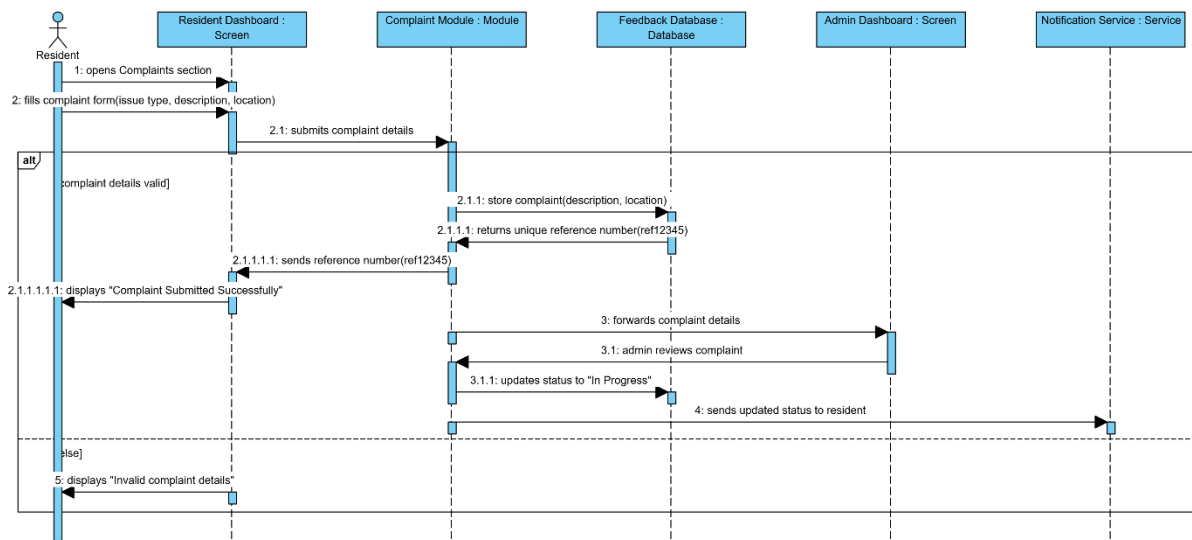
6. Admin Managing Water Truck Deliveries Sequence Diagram:



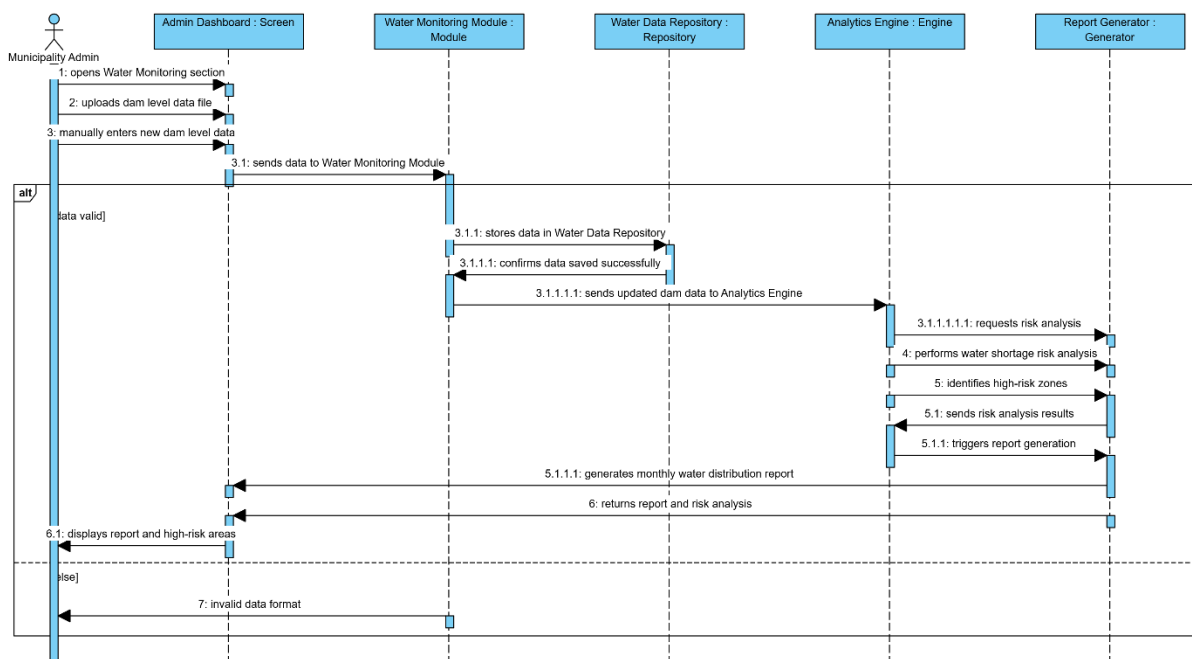
7. Resident Submitting a Complaint Sequence Diagram:



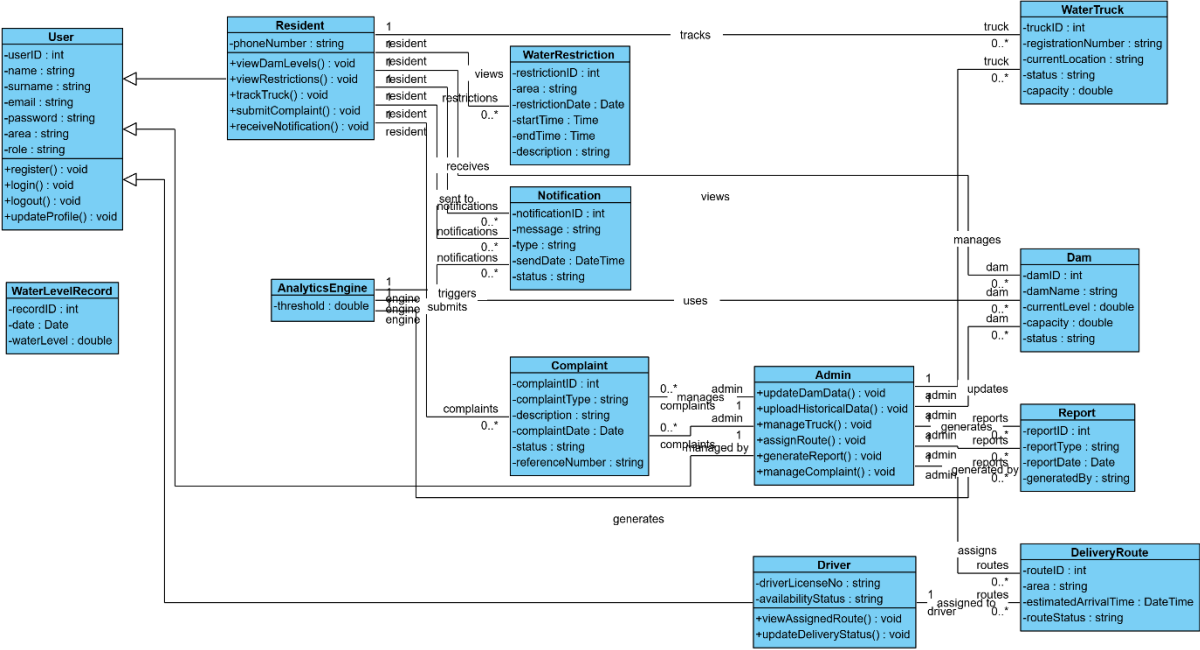
8. Admin Updating Dam Data and Generating Reports Sequence Diagram:



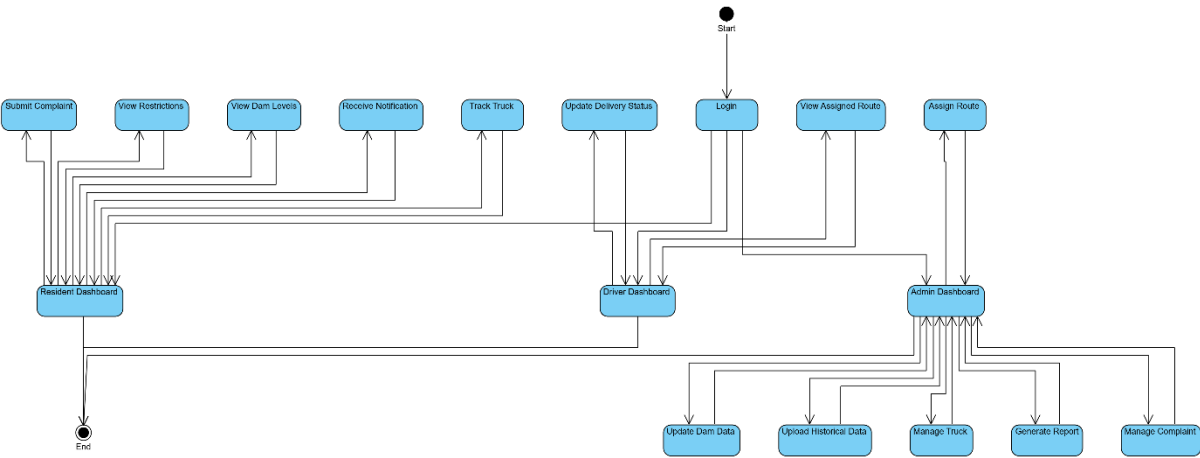
9: Admin Updating Dam Data and Generating Reports:



Class diagram



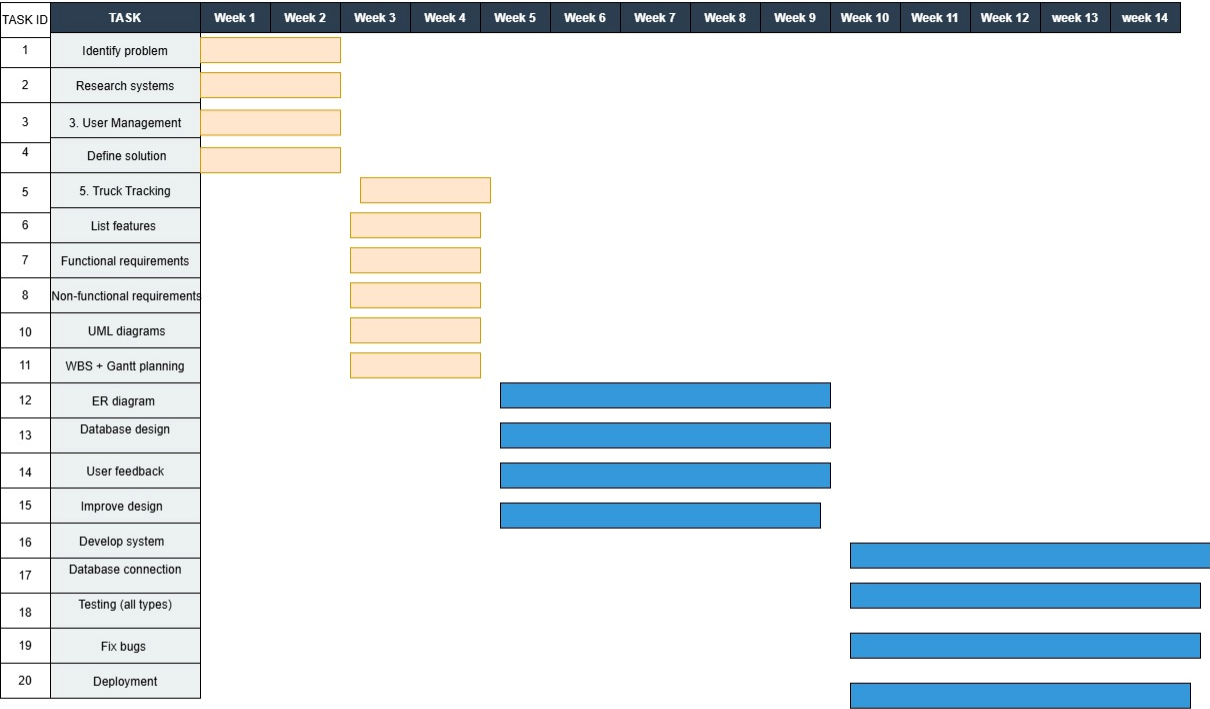
State diagram



[CLICK HERE FOR CLEAR VIEWING OF DIAGRAMS](#)

PROJECT MANAGEMENT ARTEFACTS

Water Management System - Gantt Chart



LEGEND:

=still to be done

= done

Work breakdown structure

